CONTROLLER / USB

ISOLATED POWER + I2C

ANALOG POWER / REFERENCE

J4
DISPLAY HARNESS
1 USB5V
2 USB_GND
3 HOST3V3
4 SDA_HOST
5 SCL_HOST

F1
250mA PTC
USB5V → USB5V_F

C19
10u
USB5V_F — USB_GND

C20
100n
USB5V_F — USB_GND

U7
MIE1W0505BGLVH-3R-Z
USB_GND → GND1
USB_GND → GND1
USB_GND → GND1
USB5V_F → VIN
USB5V_F → VIN
USB5V_F → EN
GND2 → AGND
GND2 → AGND
GND2 → AGND
VOUT → ISO5V
VOUT → ISO5V
VSEL → ISO5V

C21
22u
ISO5V — AGND

C22
100n
ISO5V — AGND

U9
TLV75533PDBVR
ISO5V → IN
ISO5V → EN
AGND → GND
OUT → A3V3
NC

C24
22u
A3V3 — AGND

C25
100n
A3V3 — AGND

C23
1u
ISO5V — AGND

D13
Reverse protection
ISO5V → A3V3

R37
10k
A3V3 — BIAS_SET

R38
10k
BIAS_SET — AGND

Waveshare ESP32-S3-LCD-1.28
J4: 1=5V, 2=GND, 3=3V3, 4=SDA/GPIO6, 5=SCL/GPIO7
R43/R44 DNP: host module already has 4.7k pull-ups.

U8
ISO1640BDR
HOST3V3 → VCC1
SDA_HOST → SDA1
SCL_HOST → SCL1
USB_GND → GND1
VCC2 → A3V3
SDA2 → SDA_ISO
SCL2 → SCL_ISO
GND2 → AGND

U10
OPA320AIDBVR
BIAS_SET → IN+
VBIAS → IN−
AGND → VSS
OUT → VBIAS
VDD → A3V3

U11
TLV809EA30DBZR
A3V3 → VDD
AGND → GND
RESET_N → SENSE_EN

C29
100n
A3V3 — AGND

C27
100n
A3V3 — AGND

C28
470p
BIAS_SET — AGND

R39
220R
VBIAS — AGND

C30
100n
A3V3 — AGND

R40
100k
SENSE_EN — AGND

R43
4.7k
HOST3V3 — SDA_HOST

R44
4.7k
HOST3V3 — SCL_HOST

C26
100n
HOST3V3 — USB_GND

R41
4.7k
A3V3 — SDA_ISO

R45
1k
A3V3 — AGND

R42
4.7k
A3V3 — SCL_ISO

TP13 USB5V
TP14 USB_GND
TP15 HOST3V3
TP16 ISO5V
TP17 A3V3
TP18 AGND
TP19 VBIAS
TP20 SENSE_EN
TP21 SDA_ISO
TP22 SCL_ISO

USB5V
USB5V_F
USB_GND
HOST3V3

All DUT pins remain isolated from USB ground.
Functional isolation only: not a mains/CAT-rated instrument.

Inputs 1–4
File: bank1.kicad_sch

Inputs 5–8
File: bank2.kicad_sch

Inputs 9–12
File: bank3.kicad_sch

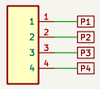
Prototype: bench validation required
12 V nominal DC | USB-powered measurement only
Power supply tester

Sheet: /
File: tester12v.kicad_sch

Title: 12 V universal pin-voltage tester – power and controller

Size: A3	Date: 2026-09-30	**Rev: B**
KiCad E.D.A. 9.0.6		Id: 1/4

INPUTS 1–4: PROTECTED SENSING AND ADC







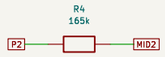














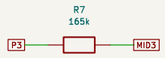




























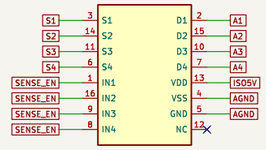


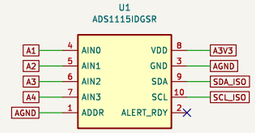
















Two 165k resistors limit input current.
D1–D12 clamp switch SOURCE pins only.
100k source bleeds permit startup with DUT connected.
Switches isolate ADC and bias when disabled.

12V nominal; 0–18V intended measurement range.
48V DC miswire target requires prototype verification.
Do not connect any terminal directly to AGND or USB_GND.

470pF filters are on the ADC side.
Vpin difference = 37.3 x ADC voltage difference (nominal).
Measure one live DC pair, other terminals open.

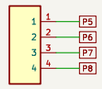
Prototype: bench validation required
12 V nominal DC | USB–powered measurement only
Power supply tester

Sheet: /Inputs 1–4/
File: bank1.kicad_sch

Title: Inputs 1–4: protected sensing and ADC

Size: A3	Date: 2026–09–30	**Rev: B**
KiCad E.D.A. 9.0.6		Id: 2/4

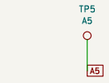
INPUTS 5–8: PROTECTED SENSING AND ADC













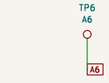














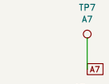


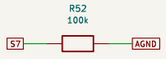






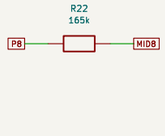


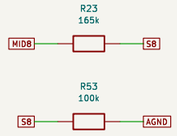


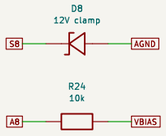




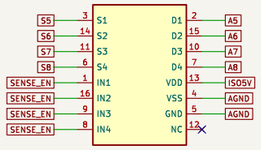


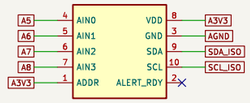
















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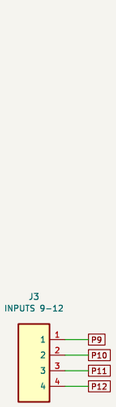
Prototype: bench validation required
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Power supply tester

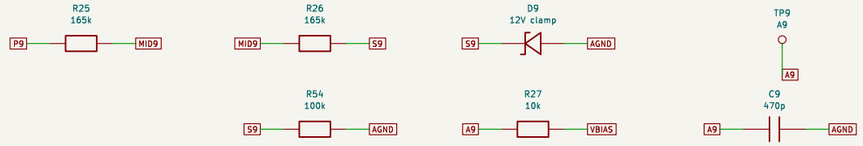
Sheet: /Inputs 5–8/
File: bank2.kicad_sch

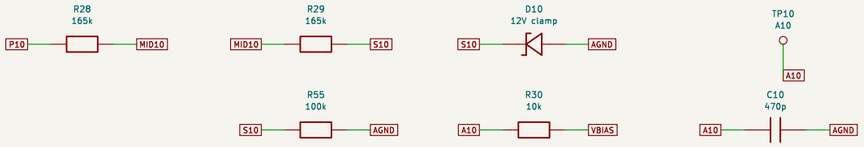
Title: Inputs 5–8: protected sensing and ADC

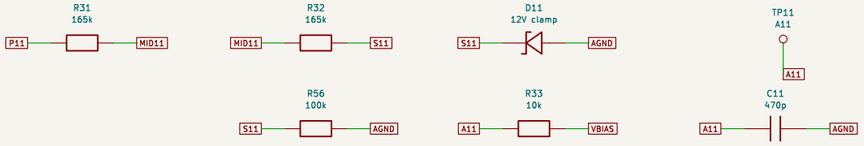
Size: A3	Date: 2026–09–30	**Rev: B**
KiCad E.D.A. 9.0.6		Id: 3/4

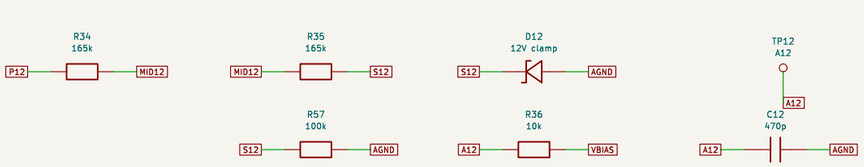
INPUTS 9–12: PROTECTED SENSING AND ADC

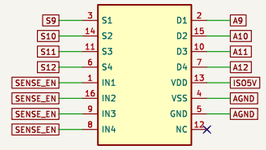


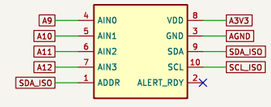
















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12 V nominal DC | USB–powered measurement only
Power supply tester

Sheet: /Inputs 9–12/
File: bank3.kicad_sch

Title: Inputs 9–12: protected sensing and ADC

Size: A3	Date: 2026–09–30	**Rev: B**
KiCad E.D.A. 9.0.6		Id: 4/4